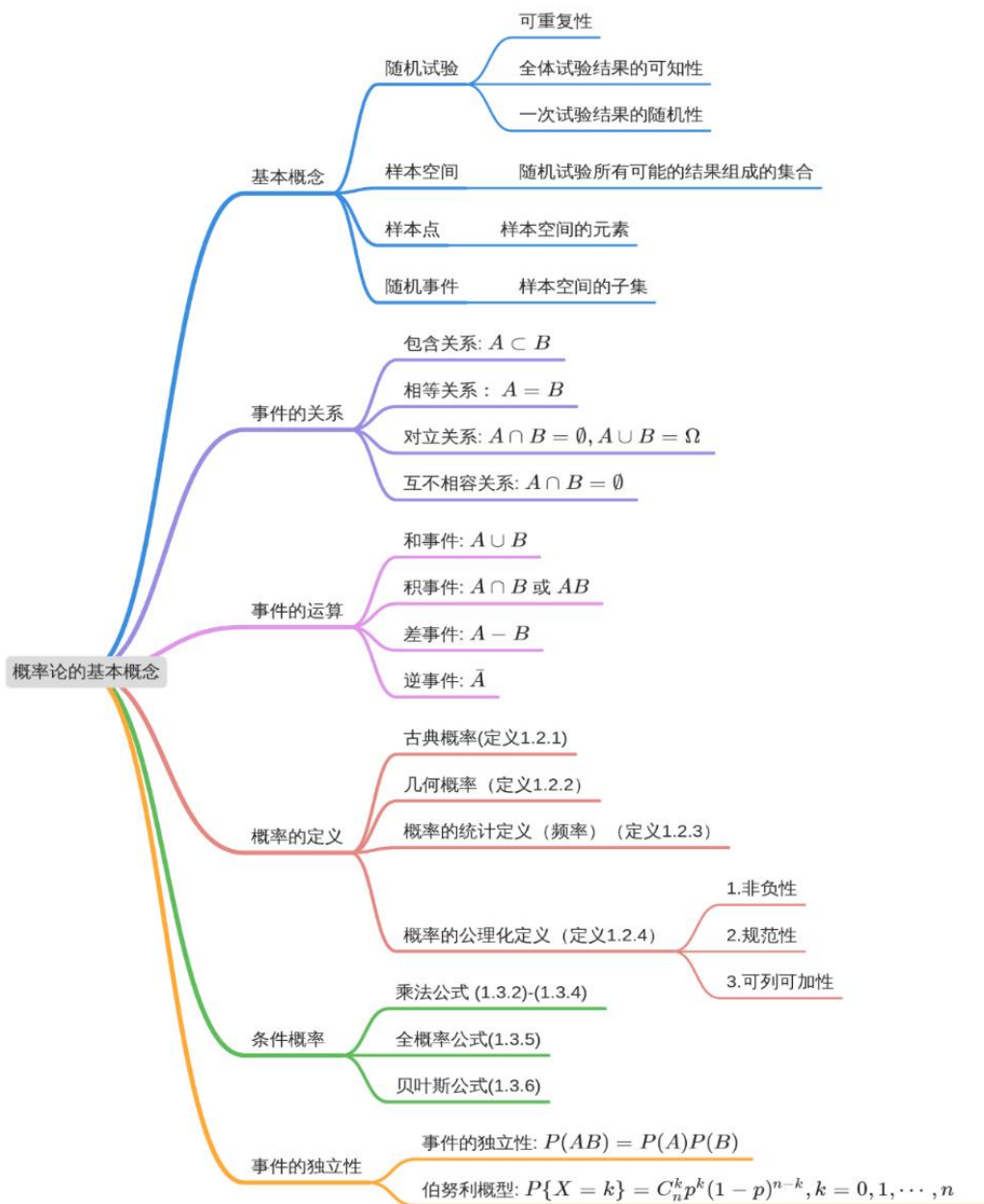




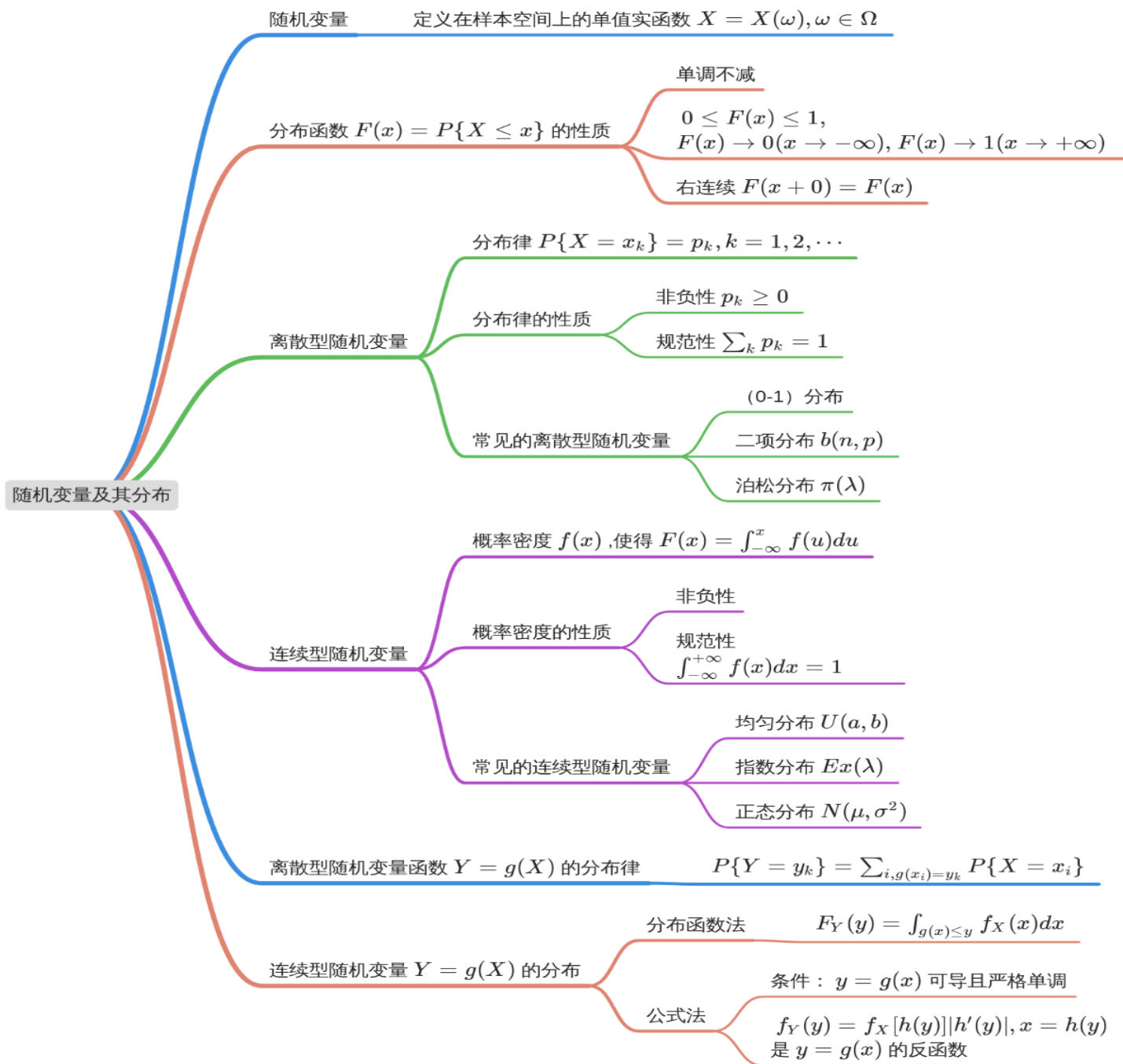
北京邮电大学数学科学学院



第一章思维导图



第二章思维导图



第三章思维导图

多维随机变量及其分布

二维随机变量的联合分布

联合分布函数 $F(x, y) = P\{X \leq x, Y \leq y\}$

(离散型) 联合分布律 $P\{X = x_i, Y = y_j\} = p_{ij}, i, j = 1, 2, \dots$

(连续型) 联合概率密度 存在 $f(x, y)$ 使得 $F(x, y) = \int_{-\infty}^y \int_{-\infty}^x f(u, v) du dv$

(X, Y) 关于 X、Y 的边缘分布

边缘分布 $F_X(x) = F(x, +\infty), F_Y(y) = F(+\infty, y)$

(离散型) 边缘分布律 $p_{i\cdot} = P\{X = x_i\} = \sum_{j=1}^{\infty} p_{ij}, p_{\cdot j} = P\{Y = y_j\} = \sum_{i=1}^{\infty} p_{ij}$

(连续型) 边缘概率密度 $f_X(x) = \int_{-\infty}^{+\infty} f(x, y) dy, f_Y(y) = \int_{-\infty}^{+\infty} f(x, y) dx$

条件分布

(离散型) 条件分布律

在 $\{Y = y_j\}$ 发生条件下 $\{X = x_i\}$ 发生的条件分布律
 $P\{X = x_i | Y = y_j\} = \frac{p_{ij}}{p_{\cdot j}}, p_{\cdot j} > 0$

(连续型) 条件概率密度

在 $Y = y$ 发生条件下的条件概率密度
 $f_{X|Y}(x|y) = \frac{f(x, y)}{f_Y(y)}, f_Y(y) > 0$

随机变量的独立性

独立性定义

$F(x, y) = F_X(x)F_Y(y)$

(离散型) 独立性

$p_{ij} = p_{i\cdot} p_{\cdot j}$

(连续型) 独立性

$f(x, y) = f_X(x)f_Y(y)$

常见的二维随机变量

均匀分布

正态分布 $N(\mu_1, \mu_2, \sigma_1^2, \sigma_2^2, \rho)$

二维正态分布的边缘分布和条件分布均为正态分布

常见的随机变量函数的分布

$M = \max\{X, Y\}$

若 X, Y 相互独立, 则 M 的分布函数为
 $F_M(x) = F_X(x)F_Y(x)$

$N = \min\{X, Y\}$

若 X, Y 相互独立, 则 N 的分布函数为
 $F_N(x) = 1 - [1 - F_X(x)][1 - F_Y(x)]$

$Z = X + Y$

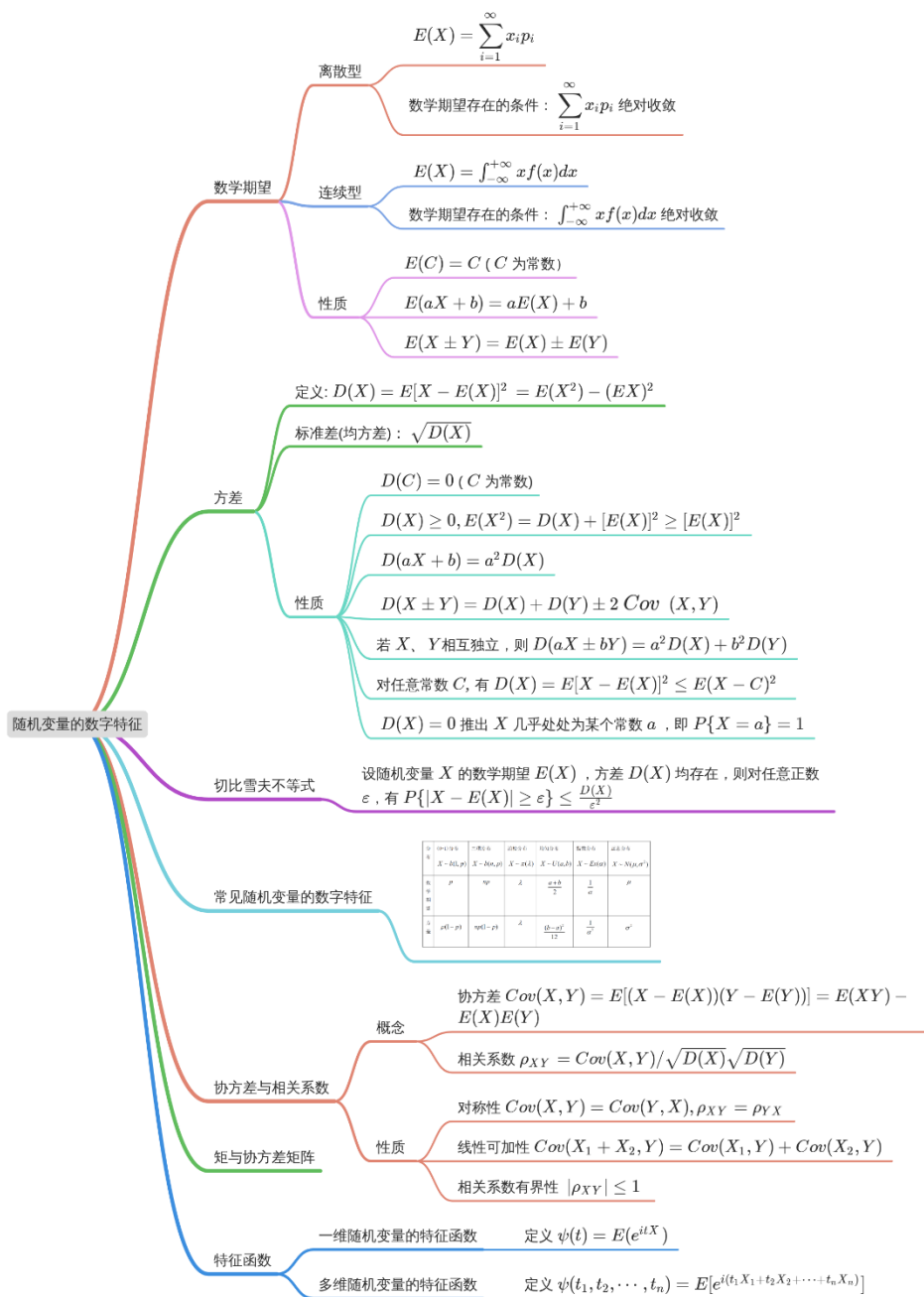
Z 的概率密度 $f_Z(z) = \int_{-\infty}^{+\infty} f(z - y, y) dy \xrightarrow{X, Y \text{ 独立}} \int_{-\infty}^{+\infty} f_X(z - y)f_Y(y) dy = f_X * f_Y(z)$

其他常见随机变量的函数

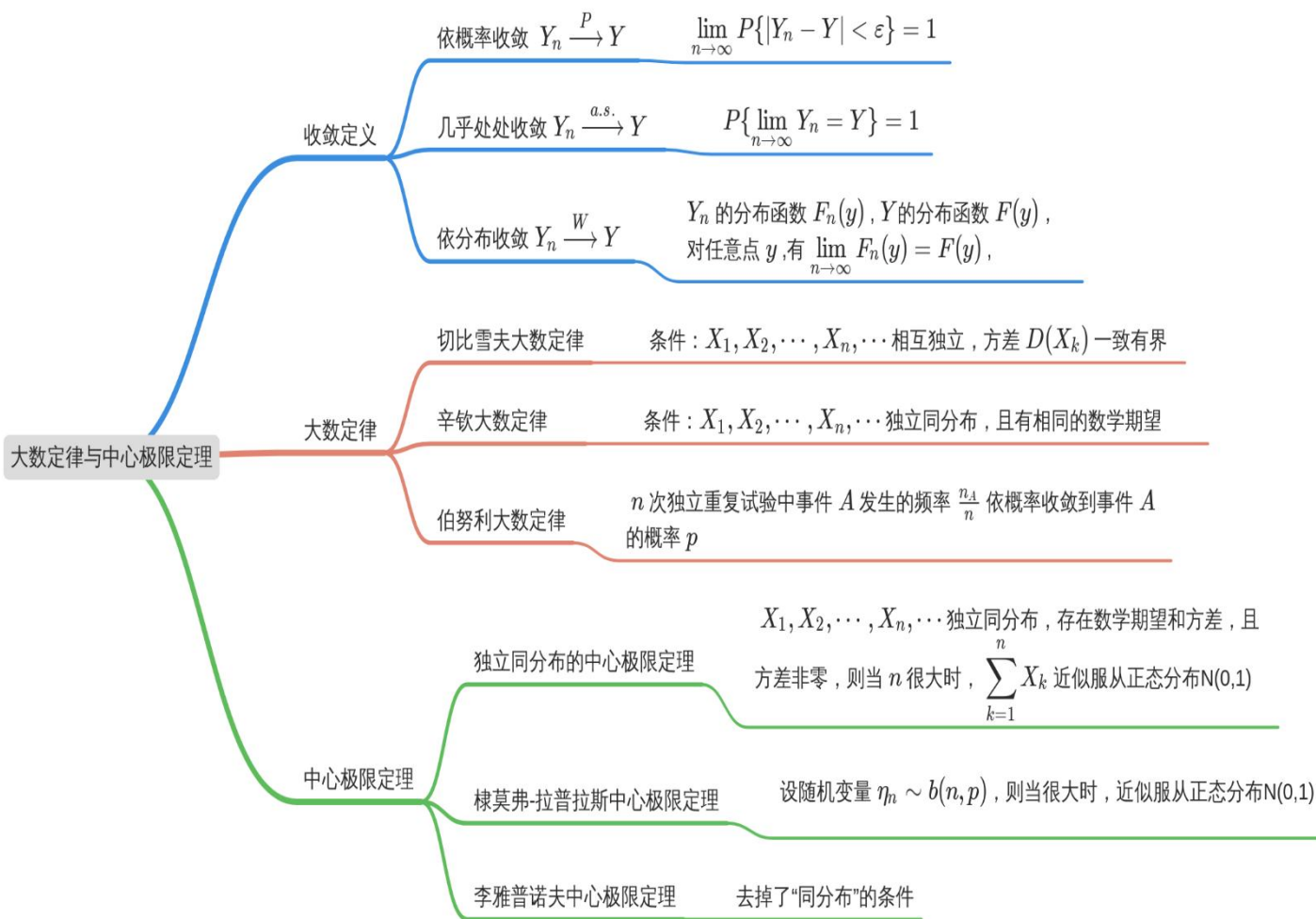
$Z = X - Y, XY, X/Y$



第四章思维导图



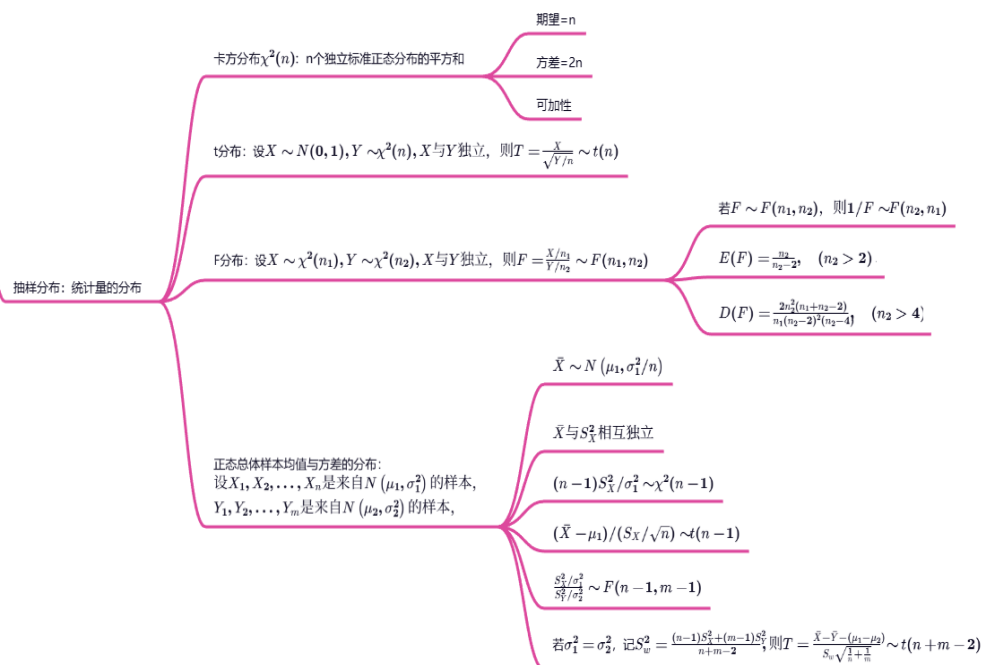
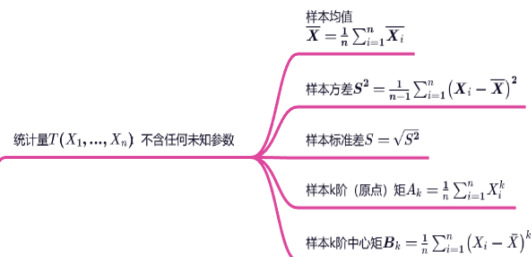
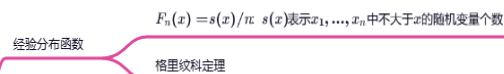
第五章思维导图



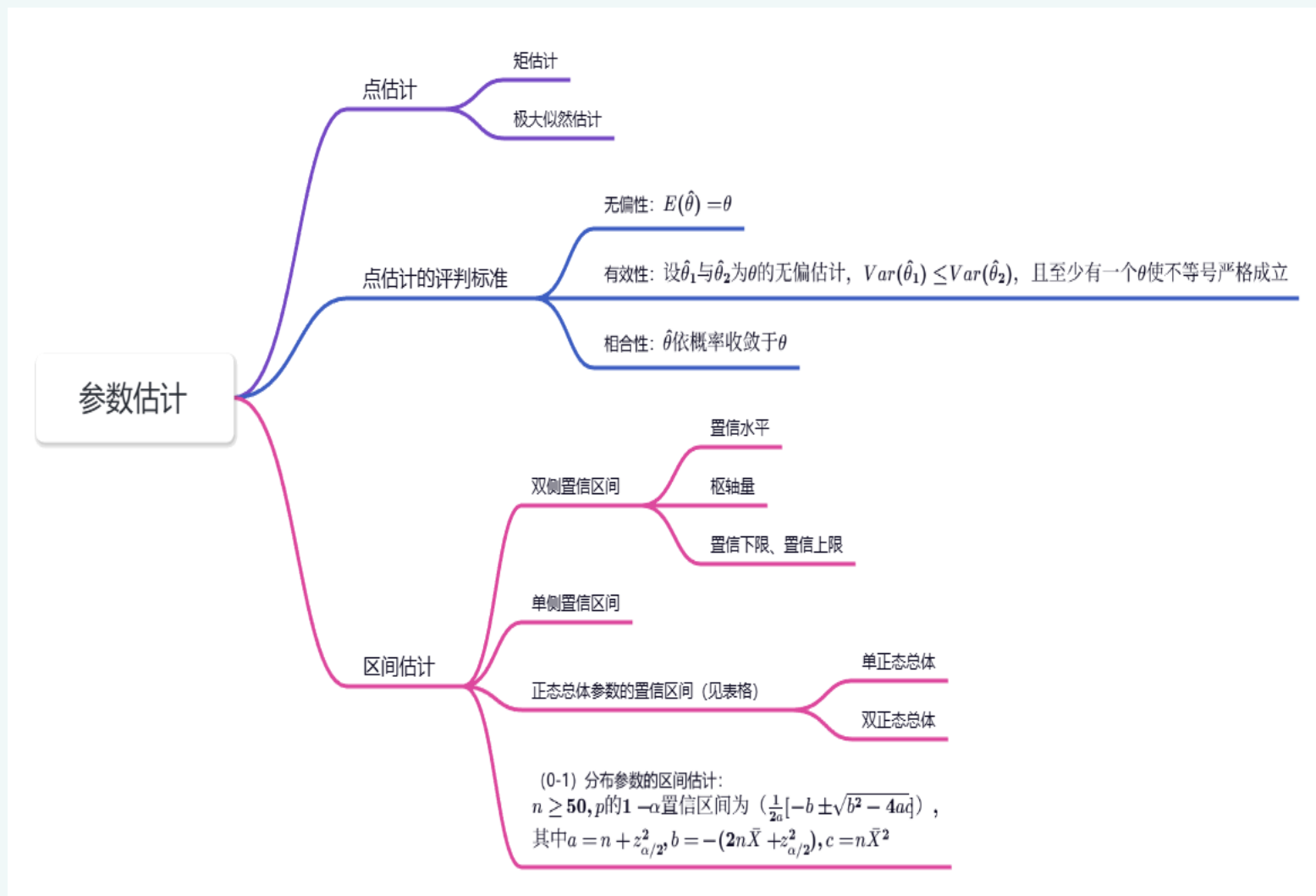
第六章思维导图



样本及抽样分布



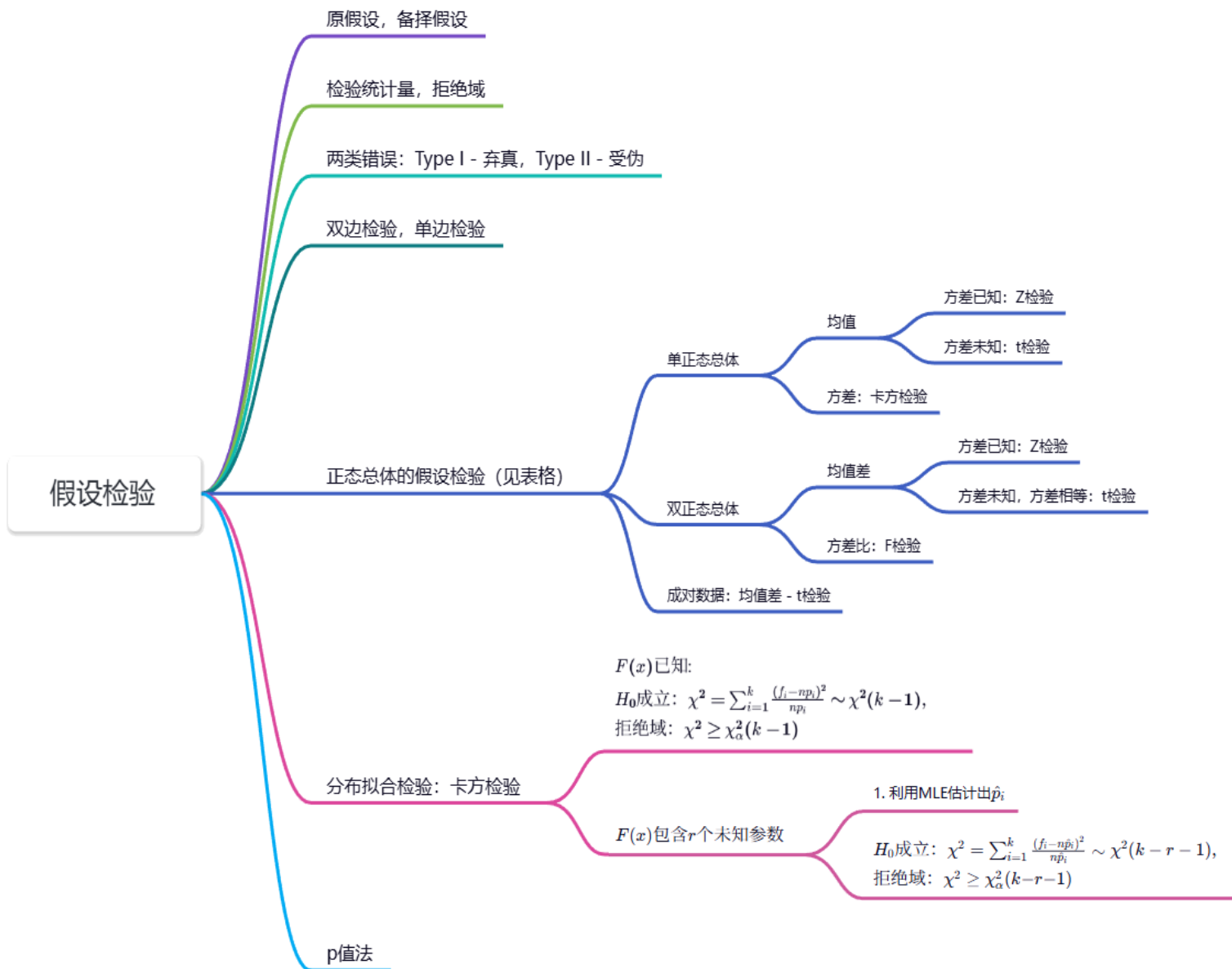
第七章思维导图



	待估参数	其他参数	W 的分布	置信区间	单侧置信限
一个正态总体	μ	σ^2 已知	$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0,1)$	$\left(\bar{X} \pm \frac{\sigma}{\sqrt{n}} Z_{\alpha/2} \right)$	$\bar{\mu} = \bar{X} + \frac{\sigma}{\sqrt{n}} Z_{\alpha}$ $\underline{\mu} = \bar{X} - \frac{\sigma}{\sqrt{n}} Z_{\alpha}$
	μ	σ^2 未知	$t = \frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t(n-1)$	$\left(\bar{X} \pm \frac{S}{\sqrt{n}} t_{\alpha/2}(n-1) \right)$	$\bar{\mu} = \bar{X} + \frac{S}{\sqrt{n}} t_{\alpha}(n-1)$ $\underline{\mu} = \bar{X} - \frac{S}{\sqrt{n}} t_{\alpha}(n-1)$
	σ^2	μ 未知	$\chi^2 = \frac{(n-1)S^2}{\sigma^2} \sim \chi^2(n-1)$	$\left(\frac{(n-1)S^2}{\chi_{\alpha/2}^2(n-1)}, \frac{(n-1)S^2}{\chi_{1-\alpha/2}^2(n-1)} \right)$	$\overline{\sigma^2} = \frac{(n-1)S^2}{\chi_{1-\alpha}^2(n-1)}$ $\underline{\sigma^2} = \frac{(n-1)S^2}{\chi_{\alpha}^2(n-1)}$
两个正态总体	$\mu_1 - \mu_2$	σ_1^2, σ_2^2 已知	$Z = \frac{(\bar{X} - \bar{Y}) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} \sim N(0,1)$	$\left((\bar{X} - \bar{Y}) \pm Z_{\alpha/2} \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}} \right)$	$\overline{\mu_1 - \mu_2} = \bar{X} - \bar{Y} + Z_{\alpha} \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$ $\underline{\mu_1 - \mu_2} = \bar{X} - \bar{Y} - Z_{\alpha} \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$
	$\mu_1 - \mu_2$	$\sigma_1^2 = \sigma_2^2 = \sigma^2$ 未知	$t = \frac{(\bar{X} - \bar{Y}) - (\mu_1 - \mu_2)}{S_w \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \sim t(n_1 + n_2 - 2)$	$\left((\bar{X} - \bar{Y}) \pm t_{\alpha/2}(n_1 + n_2 - 2) S_w \sqrt{\frac{1}{n_1} + \frac{1}{n_2}} \right)$	$\overline{\mu_1 - \mu_2} = \bar{X} - \bar{Y} + t_{\alpha}(n_1 + n_2 - 2) S_w \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$ $\underline{\mu_1 - \mu_2} = \bar{X} - \bar{Y} - t_{\alpha}(n_1 + n_2 - 2) S_w \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$
	$\frac{\sigma_1^2}{\sigma_2^2}$	μ_1, μ_2 未知	$F = \frac{S_1^2/S_2^2}{\sigma_1^2/\sigma_2^2} \sim F(n_1-1, n_2-1)$	$\left(\frac{S_1^2}{S_2^2} \frac{1}{F_{\alpha/2}(n_1-1, n_2-1)}, \frac{S_1^2}{S_2^2} \frac{1}{F_{1-\alpha/2}(n_1-1, n_2-1)} \right)$	$\left(\frac{\sigma_1^2}{\sigma_2^2} \right) = \frac{S_1^2}{S_2^2} \frac{1}{F_{1-\alpha}(n_1-1, n_2-1)}$ $\left(\frac{\sigma_1^2}{\sigma_2^2} \right) = \frac{S_1^2}{S_2^2} \frac{1}{F_{\alpha}(n_1-1, n_2-1)}$



第八章思维导图



	原假设 H_0	检验统计量	备择假设 H_1	拒绝域
1	$\mu \leq \mu_0$ $\mu \geq \mu_0$ $\mu = \mu_0$ $(\sigma^2 \text{已知})$	$Z = \frac{\bar{X} - \mu_0}{\sigma / \sqrt{n}}$	$\mu > \mu_0$ $\mu < \mu_0$ $\mu \neq \mu_0$	$z \geq z_\alpha$ $z \leq -z_\alpha$ $ z \geq z_{\alpha/2}$
2	$\mu \leq \mu_0$ $\mu \geq \mu_0$ $\mu = \mu_0$ $(\sigma^2 \text{未知})$	$t = \frac{\bar{X} - \mu_0}{S / \sqrt{n}}$	$\mu > \mu_0$ $\mu < \mu_0$ $\mu \neq \mu_0$	$t \geq t_\alpha(n-1)$ $t \leq -t_\alpha(n-1)$ $ t \geq t_{\alpha/2}(n-1)$
3	$\mu_1 - \mu_2 \leq \delta$ $\mu_1 - \mu_2 \geq \delta$ $\mu_1 - \mu_2 = \delta$ $(\sigma_1^2, \sigma_2^2 \text{已知})$	$Z = \frac{\bar{X} - \bar{Y} - \delta}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$	$\mu - \mu_0 > \delta$ $\mu - \mu_0 < \delta$ $\mu - \mu_0 \neq \delta$	$z \geq z_\alpha$ $z \leq -z_\alpha$ $ z \geq z_{\alpha/2}$
4	$\mu_1 - \mu_2 \leq \delta$ $\mu_1 - \mu_2 \geq \delta$ $\mu_1 - \mu_2 = \delta$ $(\sigma_1^2 = \sigma_2^2 = \sigma^2 \text{未知})$	$t = \frac{\bar{X} - \bar{Y} - \delta}{S_w \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$ $S_w^2 = \frac{(n_1 - 1)S_1^2 + (n_2 - 2)S_2^2}{n_1 + n_2 - 2}$	$\mu - \mu_0 > \delta$ $\mu - \mu_0 < \delta$ $\mu - \mu_0 \neq \delta$	$t \geq t_\alpha(n_1 + n_2 - 2)$ $t \leq -t_\alpha(n_1 + n_2 - 2)$ $ t \geq t_{\alpha/2}(n_1 + n_2 - 2)$

	原假设 H_0	检验统计量	备择假设 H_1	拒绝域
5	$\mu_D \leq 0$ $\mu_D \geq 0$ $\mu_D = 0$ (成对数据)	$t = \frac{\bar{D} - 0}{S_D / \sqrt{n}}$	$\mu_D > 0$ $\mu_D < 0$ $\mu_D \neq 0$	$t \geq t_\alpha(n-1)$ $t \leq -t_\alpha(n-1)$ $ t \geq t_{\alpha/2}(n-1)$
6	$\sigma^2 \leq \sigma_0^2$ $\sigma^2 \geq \sigma_0^2$ $\sigma^2 = \sigma_0^2$ (μ 未知)	$\chi^2 = \frac{(n-1)S^2}{\sigma_0^2}$	$\sigma^2 > \sigma_0^2$ $\sigma^2 < \sigma_0^2$ $\sigma^2 \neq \sigma_0^2$	$\chi^2 \geq \chi_\alpha^2(n-1)$ $\chi^2 \leq \chi_{1-\alpha}^2(n-1)$ $\chi^2 \geq \chi_{\alpha/2}^2(n-1)$ 或 $\chi^2 \leq \chi_{1-\alpha/2}^2(n-1)$
7	$\sigma_1^2 \leq \sigma_2^2$ $\sigma_1^2 \geq \sigma_2^2$ $\sigma_1^2 = \sigma_2^2$ (μ_1, μ_2 未知)	$F = \frac{S_1^2}{S_2^2}$	$\sigma_1^2 > \sigma_2^2$ $\sigma_1^2 < \sigma_2^2$ $\sigma_1^2 \neq \sigma_2^2$	$F \geq F_\alpha(n_1 - 1, n_2 - 1)$ $F \leq F_{1-\alpha}(n_1 - 1, n_2 - 1)$ $F \geq F_{\alpha/2}(n_1 - 1, n_2 - 1)$ 或 $F \leq F_{1-\alpha/2}(n_1 - 1, n_2 - 1)$

第九章思维导图

方差分析及回归分析

方差分析

单因素实验

模型: $X_{ij} = \mu + \delta_j + \epsilon_{ij}, i = 1, \dots, n_j, j = 1, \dots, s$
 $\epsilon_{ij} \sim N(0, \sigma^2), i.i.d., \sum_j n_j \delta_j = 0$

平方和分解: $S_T = S_E + S_A$

$$S_T = \sum_j \sum_i (X_{ij} - \bar{X})^2$$

$$S_E = \sum_j \sum_i (X_{ij} - \bar{X}_{\cdot j})^2$$

$$S_A = \sum_j n_j \bar{X}_{\cdot j}^2 - n \bar{X}^2$$

方差分析表

参数估计: $\hat{\mu} = \bar{X}, \hat{\mu}_j = \bar{X}_{\cdot j}, \hat{\delta}_j = \bar{X}_{\cdot j} - \bar{X}$ (若拒绝 H_0), $\hat{\sigma}^2 = S_E / (n - s)$

双因素实验

模型: $X_{ijk} = \mu + \alpha_i + \beta_j + \gamma_{ij} + \epsilon_{ijk},$
 $i = 1, \dots, r, j = 1, \dots, s, k = 1, \dots, t$
 $\epsilon_{ijk} \sim N(0, \sigma^2), i.i.d.,$
 $\sum_i \alpha_i = 0, \sum_j \beta_j = 0, \sum_i \gamma_{ij} = \sum_j \gamma_{ij} = 0$

平方和分解: $S_T = S_E + S_A + S_B + S_{AB}$

方差分析表

模型: $X_{ij} = \mu + \alpha_i + \beta_j + \epsilon_{ij},$
 $i = 1, \dots, r, j = 1, \dots, s, \epsilon_{ij} \sim N(0, \sigma^2), i.i.d.,$
 $\sum_i \alpha_i = 0, \sum_j \beta_j = 0$

平方和分解: $S_T = S_E + S_A + S_B$

方差分析表

回归分析

一元线性回归

回归模型: $Y_i = a + bx_i + \epsilon_i, \epsilon_i \sim N(0, \sigma^2), i.i.d., i = 1, \dots, n$

参数点估计

$$\hat{b} = S_{2Y} / S_{2X}$$

$$\hat{a} = \bar{Y} - \hat{b}\bar{x}$$

$$\hat{\sigma}^2 = Q_e / (n - 2)$$

假设检验

$H_0: b = 0, H_1: b \neq 0$

$$F = \frac{MS_R}{MS_e}, \text{拒绝域: } F \geq F_{\alpha}(1, n - 2), MS_R = \sum_i (\hat{Y}_i - \bar{Y})^2, MS_e = \frac{\sum_i (Y_i - \hat{Y}_i)^2}{n - 2}$$

$$T = \frac{\hat{b}}{\hat{\sigma} / \sqrt{S_{xx}}}, \text{拒绝域: } |T| \geq t_{\alpha/2}(n - 2)$$

置信区间

$$b: \left(\hat{b} \pm t_{\alpha/2}(n - 2) \frac{\hat{\sigma}}{\sqrt{S_{xx}}} \right)$$

$$\mu(x_0): \left(\hat{Y}_0 \pm t_{\alpha/2}(n - 2) \hat{\sigma} \sqrt{\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}} \right)$$

$$\text{预测区间: } \left(\hat{Y}_0 \pm t_{\alpha/2}(n - 2) \hat{\sigma} \sqrt{1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}} \right)$$

化为一元线性回归的例子

多元线性回归

回归模型: $Y = XB + \epsilon, \epsilon \sim N(0, \sigma^2 I_n)$

正规方程组: $X^T X B = X^T Y$

参数估计: $\hat{B} = (X^T X)^{-1} X^T Y$



方差来源	平方和	自由度	均方	F 比
因素 A	S_A	$s - 1$	$\bar{S}_A = \frac{S_A}{s - 1}$	$F = \bar{S}_A / \bar{S}_E$
误差	S_E	$n - s$	$\bar{S}_E = \frac{S_E}{n - s}$	
总和	S_T	$n - 1$		

$H_0: \delta_1 = \delta_2 = \cdots = \delta_s = 0, H_1: \delta_j$ 不全为零, $j = 1, 2, \dots, s.$

拒绝域: $F \geq F_\alpha(s - 1, n - s)$



方差来源	平方和	自由度	均方	F 比
因素 A	S_A	$r-1$	$\bar{S}_A = \frac{S_A}{r-1}$	$F_A = \frac{\bar{S}_A}{\bar{S}_E}$
因素 B	S_B	$s-1$	$\bar{S}_B = \frac{S_B}{s-1}$	$F_B = \frac{\bar{S}_B}{\bar{S}_E}$
交互作用	$S_{A \times B}$	$(r-1)(s-1)$	$\bar{S}_{A \times B} = \frac{S_{A \times B}}{(r-1)(s-1)}$	$F_{A \times B} = \frac{\bar{S}_{A \times B}}{\bar{S}_E}$
误差	S_E	$rs(t-1)$	$\bar{S}_E = \frac{S_E}{rs(t-1)}$	
总和	S_T	$rst-1$		



$H_{01}:\alpha_1 = \alpha_2 = \cdots = \alpha_r = 0, H_{11}:\alpha_i$ 不全为零.
 拒绝域: $F \geq F_\alpha(r-1, rs(t-1))$

$H_{02}:\beta_1 = \beta_2 = \cdots = \beta_s = 0, H_{12}:\beta_j$ 不全为零.
 拒绝域: $F \geq F_\alpha(s-1, rs(t-1))$

$H_{03}:\gamma_{11} = \gamma_{12} = \cdots = \gamma_{rs} = 0, H_{13}:\gamma_{ij}$ 不全为零.
 拒绝域: $F \geq F_\alpha((r-1)(s-1), rs(t-1))$ ¹⁵

方差来源	平方和	自由度	均方	F 比
因素 A	S_A	$r-1$	$\bar{S}_A = \frac{S_A}{r-1}$	$F_A = \frac{\bar{S}_A}{\bar{S}_E}$
因素 B	S_B	$s-1$	$\bar{S}_B = \frac{S_B}{s-1}$	$F_B = \frac{\bar{S}_B}{\bar{S}_E}$
误差	S_E	$(r-1) \times (s-1)$	$\bar{S}_E = \frac{S_E}{(r-1)(s-1)}$	
总和	S_T	$rs-1$		

$H_{01}: \alpha_1 = \alpha_2 = \cdots = \alpha_r = 0$, $H_{11}: \alpha_i$ 不全为零. 拒绝域: $F \geq F_\alpha(r-1, (r-1)(s-1))$

$H_{02}: \beta_1 = \beta_2 = \cdots = \beta_s = 0$, $H_{12}: \beta_j$ 不全为零. 拒绝域: $F \geq F_\alpha(s-1, (r-1)(s-1))$



方差来源	平方和	自由度	均方	F 比
回归	$S_R = \sum_{i=1}^n (\hat{y}_i - \bar{y})^2 = \hat{b}^2 S_{xx}$	$f_R = 1$	$MS_R = \frac{S_R}{f_R}$	$F = \frac{MS_R}{MS_e}$
残差	$S_e = \sum_{i=1}^n (y_i - \hat{y}_i)^2 = S_T - S_R$	$f_e = n - 2$	$MS_e = \frac{S_e}{f_e}$	
总和	$S_T = \sum_{i=1}^n (y_i - \bar{y})^2 = S_{yy}$	$f_T = n - 1$		

$H_0: b = 0, H_1: b \neq 0$. 拒绝域: $F \geq F_\alpha(1, n - 2)$

